

Project Name – Email Automation

- **Project Overview:**

The Email Automation System is a backend application that automates the process of sending emails to multiple recipients. It allows users to upload a CSV file containing recipient email addresses, compose emails, schedule email delivery, use reusable templates, manage attachments, track email status, and handle email replies. The system also maintains email logs and follows email authentication practices to reduce the chances of emails being marked as spam.

- **Project Objectives:**

- Automate the process of sending emails.
- Reduce manual effort in sending bulk emails.
- Improve communication efficiency.
- Support scheduled email delivery.
- Maintain logs of all email activities.
- Ensure reliable and secure email delivery.

- **Functional Requirements**

1. The system should authenticate users using login credentials.
2. The system should read email addresses from a CSV file, validate them, and send emails to all valid recipients.
3. The system should automatically check the uploaded CSV file and remove any duplicate email addresses so nobody gets the same email twice.
4. Users should be able to enter the email subject and message.
5. Users should be able to schedule emails to be sent later, and the system should automatically send them at the specified time.
6. Users should be able to attach files such as PDF, Word, Excel, or image files.
7. The system should show whether each email was sent successfully or failed. If sending fails due to a temporary problem, the system should try again automatically.
8. The system should allow users to create and save email templates with placeholders like **{name}** and **{date}**, which are replaced with actual values when the email is sent.
9. The system should maintain a log of all sent, received, and failed emails, along with their status, date, and time.
10. The system should be able to read and handle replies received for sent emails.
11. The system should support sending and receiving emails through multiple email service providers, such as Gmail, Outlook, and Yahoo Mail.

12. The system should detect and handle errors such as invalid email addresses, failed email delivery, and internet connectivity issues, and display appropriate error messages to the user.

Test cases:

TC ID	Test Case	Example/Test Data	Expected Result
TC_001	Verify that the system allows users to log in with valid credentials.	Username: admin@gmail.com Password: Admin@123	User is logged in successfully and redirected to the dashboard.
TC_002	Verify that the system displays an error for invalid login credentials.	Username: admin@gmail.com Password: Wrong123	Error message "Invalid username or password" is displayed.
TC_003	Verify that the system reads and validates email addresses from a CSV file.	CSV contains john@gmail.com, abc@yahoo.com	Valid email addresses are accepted for sending.
TC_004	Verify that invalid email addresses are rejected.	CSV contains john@gmail, abc@.com, test123	Invalid email addresses are identified and skipped with an error message.
TC_005	Verify that duplicate email addresses are removed.	CSV contains john@gmail.com, john@gmail.com, abc@yahoo.com	Duplicate john@gmail.com is removed and email is sent only once.

TC ID	Test Case	Example/Test Data	Expected Result
TC_006	Verify that users can enter the email subject and message.	Subject: Meeting Reminder Message: Please attend the meeting at 10 AM tomorrow.	Subject and message are accepted successfully.
TC_007	Verify that users can schedule an email.	Schedule Date: 25-06-2026 Time: 10:00 AM	Email is sent automatically on 25-06-2026 at 10:00 AM.
TC_008	Verify that scheduling with a past date is not allowed.	Schedule Date: 20-06-2026	System displays "Invalid schedule date/time."
TC_009	Verify that supported file types can be attached.	Attach Report.pdf or Invoice.xlsx	File is attached successfully.
TC_010	Verify that unsupported files are rejected.	Attach virus.exe	System displays "Unsupported file type."
TC_011	Verify that email status is displayed after sending.	Send email to john@gmail.com	Status shows Sent Successfully .
TC_012	Verify retry mechanism on temporary failure.	Disconnect internet while sending	System retries automatically after connection is restored.
TC_013	Verify that users can	Template Name: Welcome Mail	Template is saved successfully.

TC ID	Test Case	Example/Test Data	Expected Result
	save email templates.	Content: Hello {name}, Welcome!	
TC_014	Verify placeholder replacement.	Template: Hello {name} CSV Name: John	Email sent as Hello John.
TC_015	Verify that sent emails are logged.	Send email to john@gmail.com	Log shows recipient, status Sent , date and time.
TC_016	Verify that failed emails are logged.	Send email to abc@invalid	Log shows Failed with reason "Invalid Email Address".
TC_017	Verify that replies are received and displayed.	Reply received from john@gmail.com	Reply appears in the inbox/reply section.
TC_018	Verify spam prevention mechanisms.	Send authenticated email using SMTP with SPF/DKIM	Email is delivered with a lower chance of being marked as spam.
TC_019	Verify Gmail support.	Configure Gmail account and send email	Email is sent successfully through Gmail.
TC_020	Verify Outlook support.	Configure Outlook account and send email	Email is sent successfully through Outlook.
TC_021	Verify Yahoo Mail support.	Configure Yahoo Mail account and send email	Email is sent successfully through Yahoo Mail.

TC ID	Test Case	Example/Test Data	Expected Result
TC_022	Verify handling of invalid email addresses.	Recipient: abcgmail.com	System displays "Invalid email address."
TC_023	Verify handling of internet connectivity issues.	Turn off Wi-Fi while sending email	System displays "No Internet Connection" and retries later.
TC_024	Verify sending with an empty CSV file.	Upload an empty CSV	System displays "No recipients found."
TC_025	Verify handling of mixed valid and invalid email addresses.	CSV: john@gmail.com, abc@.com, alice@yahoo.com	Emails are sent only to john@gmail.com and alice@yahoo.com; invalid email is skipped.
TC_026	Verify processing of a large CSV file.	CSV containing 10,000 email addresses	System processes the file without crashing and sends emails successfully.
TC_027	Verify viewing email logs.	Open Email Logs page	Sent, failed, and received emails are displayed with date, time, and status.
TC_028	Verify editing an existing email template.	Change Welcome Mail to Welcome to ABC Company	Updated template is saved successfully.

TC ID	Test Case	Example/Test Data	Expected Result
TC_029	Verify attachment size validation.	Attach a 30 MB PDF when limit is 25 MB	System displays "File size exceeds the maximum limit."
TC_030	Verify sending an email without a subject.	Subject: <i>(Blank)</i> Message: Hello Team	System displays "Subject cannot be empty" (or sends if blank subjects are allowed, based on requirements).

- **Acceptance Criteria**

1. Login
2. CSV upload and validation
3. Duplicate removal
4. Email scheduling
5. Email templates
6. Email status and retry
7. Reply handling
8. Error handling

- **Future Scope:**

1. The system should minimize the chances of emails being marked as spam by using proper email authentication and sending practices